Constraint Transport and Temporally Gated Expression Lineage in a Finite Local Iota System

Douglas Clark

Working draft

Abstract

Information is investigated as an interaction-induced change in reachable future states. For uniform laws on nested reachable supports, the logarithmic support-ratio measure is exactly a Kullback–Leibler relative entropy. This identity motivates a policy- and horizon-indexed future-law divergence and a prospective bridge to the geometric quantum relative entropy used in Gravity from Entropy (GfE). A finite local lattice system is used to test whether relational constraints can persist when the local cells realizing them are replaced. Three strict constraint-transport events show tracked regions disconnecting after bridge loss and reconnecting through a causally necessary disjoint bridge under payload-free dynamics. Prospective ancestry then shows expression lineage surviving stationary-address death. In 192 fresh branches, one-decision interventions reopened all 36 targeted successor routes; all six gate interventions produced immediate children and 28 of 30 exhaustion interventions eventually handed off. A complete screen of 16 unused initialization seeds froze two new geometries before terminal outcomes; the same protocol rescued all eight new gate-failure cases and produced directionally correct exhaustion and preemption mediators. The result is bounded causal evidence that a lawful successor must remain available and be realized before closure. A separately committed reverse intervention vetoed 164 natural unique-child events: 122 branches realized an alternate child and handed off, while 42 realized none and lost the lineage. Extending those branches through prospectively frozen sequential vetoes produced an observed ordered cut in 143 of 164 branches by depth eight; 19 remained above the protocol maximum and 2 retained the original source at the fixed horizon. A preregistered independent replication then screened new initializations before outcomes, selected 4 geometries, and repeated the policy on 79 factual continuations. All four geometries retained a continuation after the first veto; 68 branches cut by depth eight, 10 remained above the cap, and 1 was source-survival censored. These are bounded simulator-level causal results, not a globally minimal cut or a claim of physical correspondence. The distributional diagnostic and GfE mapping are formal research proposals; neither is tested by the reported events.

1 Organizing Hypothesis

The organizing hypothesis of this research program is:

Information is the change in the reachable caused by interaction.

This is a hypothesis to be operationalized and tested, not an established law of physics. It motivates a narrower experimental question: can finite local interactions produce constraints on reachable futures that persist even when the local state realizing those constraints is replaced?

The primitive experimental target is therefore the latent counterfactual structure of lawful successor events, rather than an early analogue of mass, energy, force, or a classical field. The lattice

stores state; a marked relational transition graph represents which successor interactions remain possible; and coherence is provisionally the organization that preserves and transmits part of that structure across realized transitions. Familiar physical observables, if they emerge, are downstream quantities to explain rather than primitive variables to imitate. Accordingly, a proposed experiment must state which aspect of the successor structure it causally changes and measures.

For a state s_t, define the locally reachable next-state set as

$$\mathcal{R}_x(t+1 \mid s_t) = \left\{(A', E') : (A', E') \text{ is admissible under } \mathcal{L}\right\}. \tag{1}$$

When interaction strictly reduces the reachable set, a cardinality measure is

$$\Delta I_x = \log \frac{|\mathcal{R}_x^{\text{prior}}(t+1)|}{|\mathcal{R}_x^{\text{posterior}}(t+1)|}. \tag{2}$$

Equation 2 does not distinguish all set reshapings that preserve cardinality. A set- or distribution-sensitive extension remains open.

1.1 A distribution-sensitive constraint diagnostic

Let $P_{x,\tau}^{\text{prior},\pi}$ and $P_{x,\tau}^{\text{post},\pi}$ be probability laws for the same declared future observable $F_{t:t+\tau}$ under policy or intervention π, before and after the focal interaction. The outcome space, measure, boundary, coarse-graining, resources, and horizon are held fixed. A candidate distribution-sensitive diagnostic is

$$\mathfrak{C}_{x,\tau}^{\pi} := D_{\text{KL}}\left(P_{x,\tau}^{\text{post},\pi} \middle\| P_{x,\tau}^{\text{prior},\pi}\right). \tag{3}$$

Equation 3 requires $P_{x,\tau}^{\text{post},\pi} \ll P_{x,\tau}^{\text{prior},\pi}$; otherwise it may diverge. It is directional, reference-relative, and indexed by a declared future task. It is an interventional constraint diagnostic, not a new definition of information or of the Information–Constraint Principle.

The existing support-ratio measure is an exact special case. If $\mathcal{R}^{\text{post}} \subseteq \mathcal{R}^{\text{prior}}$ and U_{post}, U_{prior} are uniform laws on those supports, then

$$\begin{aligned} D_{\text{KL}}(U_{\text{post}} \| U_{\text{prior}}) &= \sum_{r \in \mathcal{R}^{\text{post}}} \frac{1}{|\mathcal{R}^{\text{post}}|} \log \frac{1/|\mathcal{R}^{\text{post}}|}{1/|\mathcal{R}^{\text{prior}}|} \\ &= \log \frac{|\mathcal{R}^{\text{prior}}|}{|\mathcal{R}^{\text{post}}|} = \Delta I_x. \end{aligned} \tag{4}$$

The orientation is essential: reversing the arguments diverges for strict support pruning because the posterior law assigns zero probability to some prior alternatives. More generally, a scalar divergence can register soft reweighting but still suppresses the identity and relational organization of the changed futures. The complete indexed future-law family therefore remains the primary object; any aggregation over policies must be declared [12].

1.2 Relation to prior work

The logarithm of a finite number of alternatives is Hartley information [7]; therefore Equation 2 is a difference of Hartley support measures when its alternatives are treated uniformly, not a new information measure. Restriction-centered accounts have also identified information with limitations

on possible variable values [18]. Cause–effect repertoires and effective information likewise quantify how mechanisms or interventions constrain possible states in discrete dynamical systems [1, 9].

Counterfactual approaches to semantic information test whether scrambling or erasure changes a system's future or viability [2, 11, 15]. Persistence of organization through component turnover is central to autopoiesis and constraint-closure accounts [3, 13, 16]. Related operational tools include local causal states for coherent structures, reachability in temporal networks, and causal ancestry in cellular automata [8, 10, 14].

At the physical-foundations level, constructor theory formulates information in terms of possible and impossible transformations [6]. A recent preprint also describes information and computation through persistent relational constraints on admissible transitions [17]. Gravity from Entropy (GfE) instead proposes a continuum gravitational action built from a geometric quantum relative entropy between a spacetime metric and a metric induced by matter and curvature [4, 5]. Its operands are metric operators, not reachable-future distributions; the possible bridge is addressed in Section 5.1. The present work therefore does not claim novelty for information-as-restriction, organization across turnover, or causal intervention on future possibilities. Its narrower novelty target is the exact operational combination of disjoint bridge replacement, paired stochastic intervention, and ordered realization cuts in this finite local system. Whether that diagnostic is new and general remains to be tested against the cited methods and matched controls.

The experiments also distinguish a state-level possibility from its lifetime under ordered resolution. Let $\mathcal{R}^{(k)}$ denote the routes still admissible after event position k within a declared update. A route may be lawful in $\mathcal{R}^{(k)}$ yet disappear from $\mathcal{R}^{(k+1)}$ when a competing lawful event exhausts its source or preempts its target. This notation does not assert non-Markovian physics: the complete simulator state, RNG state, and scheduler determine the update. It makes the causal question operational: must an ancestry-preserving successor be realized before its route closes?

2 Provisional Model

At permanent lattice address x, write an iota provisionally as

$$I_x(t) = (x, A_x(t), E_x(t)) \,, \tag{5}$$

where $A_x(t)$ is its attribute state and $E_x(t)$ is its expression state. The 26-cell Moore neighborhood is denoted by $N_{26}(x,t)$. A fixed local law $\mathcal{L}$ defines the generative kernel

$$(A_x(t+1), E_x(t+1)) = \mathcal{L}\left(A_x(t), E_x(t), N_{26}(x,t)\right). \tag{6}$$

Equation 6 is a model definition. No identification with a fundamental physical particle is assumed.

2.1 Operational vocabulary

Iota (I_x(t)) A stationary lattice address together with the local attribute and expression state instantiated at that address.

Reachable set (\mathcal{R}_x(t+1\mid s_t)) The set of next local states admissible from state s_t under the declared local law.

Interaction-induced information (\Delta I_x) The change in the reachable caused by interaction. For strict reachable-set reduction, the current cardinality measure is the logarithm of the prior-to-posterior reachable-set size ratio.

Coherence A connected active structure measured using the explicitly declared neighborhood connectivity. It is not assumed to be a fundamental physical object.

Bridge (B) A set of active lattice cells whose removal changes reachability between tracked regions in the audited state.

Strict constraint transport Loss and restoration of reachability between the same tracked predecessor regions through a causally necessary successor bridge disjoint from the predecessor bridge.

Payload-free (P=0) World.P is zero at every observed offset in the audited continuation.

Ledger exactness The experiment's conservation ledger has zero recorded discrepancy at every audited offset.

Predecessor address lineage A conservative lineage that remains alive while at least one original stationary iota address of the predecessor region remains active; later reactivation after complete address loss does not restore the terminated lineage.

Prospective expression lineage A non-resurrecting lineage propagated only when a successful transfer from a current lineage address newly activates its target in that tick; retained active addresses remain members, already-active targets do not inherit identity, and ambiguous dual claims invalidate a strict result.

Expression-handoff susceptibility (p_{\mathrm{handoff}}) Among continuation branches from a fixed state in which the original singleton address becomes inactive within the declared horizon, the fraction retaining a nonempty, collision-free prospective singleton expression lineage immediately after that first deactivation.

Candidate lineage-continuation rule A frozen mapping from features available immediately before original-address deactivation to the presence or absence of a nonempty prospective expression lineage immediately afterward; a predictive marker is not treated as a primitive law unless it is both sufficiently sensitive and independently replicated.

Temporal continuation gate An ordered event decision whose resolution before source termination changes whether an ancestry-preserving successor is realized; it is identified by a frozen intervention that changes that decision while leaving the pre-intervention state and consumed random draw fixed.

Opportunity lifetime (\tau_{\mathrm{opp}}) The ordered event or tick interval during which a specified successor transition remains admissible, ending when that transition is realized or a declared lawful event makes the route unavailable.

Realization redundancy For a factual ancestry-producing event selected before intervention, temporal redundancy occurs when vetoing that event leads to another ancestry-preserving child within the declared follow-up. The selected event is branch-necessary when the address instead dies without inherited continuation; continued source activity at the horizon is censored.

Sequential realization cut depth (D_{\mathrm{seq}}) Under a frozen retained-amplitude veto policy and fixed follow-up, the smallest depth d along a branch's prospectively committed selector chain at which vetoing the first d ancestry-producing events causes the original

address to deactivate without a live collision-free descendant. Handoff after the administrative maximum is right-censored above that depth; continued original-address activity at the horizon is source-survival censored. The depth is prefix-minimal for this ordered policy, not a graph-theoretic minimum over arbitrary intervention sets.

Latent counterfactual structure The marked relational structure of lawful successor events still realizable from a declared complete state under the declared law, including their conditional order, compatibility, and closure. It is not the single factual trajectory, a scalar field, or a claim that every successor is simultaneously available.

Frozen rooted graph-role signature For Finding 075, the tuple of face6 and face18 active-neighbor counts, face6 and cube26 neighbor-component counts, and the sorted induced cube26-degree sequence among a target's active cube26 neighbors, conditioned on exact cube26 shell-one and Chebyshev shell-two occupancy. Equality of this signature is a declared matching representation, not graph isomorphism or an established sufficient statistic for successor dynamics.

Coherence-conditioned transition geometry The organized, potentially anisotropic and history-dependent structure of lawful next transitions conditioned on a focal coherence, its marked relational neighborhood, and its recent causal history. The term does not presuppose a scalar, vector, tensor, Euclidean field, force, gravity analogue, or independently confirmed predictive representation.

Successor openness (O_{\mathrm{succ}}) For a declared pre-step state and radius-two basin, the sum of frozen transfer probabilities over tick-start-eligible inactive-target activation edges in the marked hard-transfer candidate graph, accompanied by the number of distinct activation targets. It is not an entropy, an exhaustive multi-tick reachable set, or an independently confirmed predictor.

Repair affordance (A_{\mathrm{repair}}) For a declared pre-step state and radius-two basin, the sum of frozen transfer probabilities over eligible depleted-target secondary edges classified as component_bridge_cube26 or local_bridge_face6, accompanied by distinct target and source counts. It measures one-step bridge/reclosure accessibility under the declared representation, not realized repair or a universal counterfactual quantity.

Factorial interaction (I_{T\times M}) For natural, topology-only, marks-only, and joint states, the declared instrument contrast joint minus topology-only minus marks-only plus natural. A nonzero value establishes non-additivity of the instrument under those interventions; by itself it does not establish a downstream causal or predictive effect.

3 Constraint-Transport Test

Finding 060 replayed the 36 Class B bridge-relocation events identified by the preceding class-matrix experiment. The audit required a role-bearing predecessor bridge and tracked the exact predecessor regions through bridge loss and reconnection. Pairwise reachability and shortest-path accessibility were measured before loss, during the split, and after rebridge. The successor bridge was then removed to test whether it was causally necessary.

Let $C_{UV}(t)$ equal one when tracked regions U_t and V_t are mutually reachable and zero otherwise. A strict event requires

$$C_{UV}(t_{\mathrm{pre}}) = 1, \qquad C_{UV}(t_{\mathrm{split}}) = 0, \qquad C_{UV}(t_{\mathrm{post}}) = 1, \tag{7}$$

together with disjoint predecessor and successor bridges, $B_{\mathrm{pre}} \cap B_{\mathrm{post}} = \varnothing$, and loss of restored reachability after removing B_{post}.

The analysis code and exact source tables are linked from Section 7.

3.1 Expression ancestry and terminal-event intervention

A prospective expression lineage retains an active address and inherits a newly activated target only from a successful transfer whose source belonged to that lineage before the update. Already-active targets do not inherit identity, terminated lineages do not resurrect, and competing claims to one child mark a collision.

For Finding 067, a default-off observer recorded selected transfer candidates in evaluation order, including target state, transfer probability, random draw, and effective decision. Paired observer-free and sham-intervention worlds were required to remain exactly equal in every state array and RNG state. Each causal replay consumed the natural random draw but changed exactly one decision: permit a failed inactive-successor gate, veto the non-child transfer spending the source's final amplitude, or veto the earlier activation preempting a target. The primary mediators were route reopening and immediate inherited-child creation; eventual handoff at first address death was evaluated separately.

3.2 Independent-geometry holdout

The independent screen exhausted a frozen 16×8 matrix: unused initialization seeds 57–72 and eight distinct continuation streams per seed. Every Class B candidate received the unchanged strict transport audit. Before terminal outcomes were examined, the source manifest retained at most the lexicographically first eligible singleton transport per initialization seed. Each selected source was then branched 32 times at rebridge, rather than at a later terminal-aligned state, and evaluated with the same intervention policy. Initialization seed is the independent geometry unit; branch outcomes are conditional stochastic replicates.

3.3 Reverse necessity and redundancy test

Finding 068 separated factual selection from intervention. First, all 256 fresh factual branches were completed and every collision-free, single-address, natural child event was written to a selector manifest. The manifest stored the candidate and gate ordinals, source, target, amplitude, target state, probability, and random draw and was committed before replay. Every selected factual event had exactly one inherited child.

The replay stage accepted only the committed manifest hash. It reproduced an exact sham, matched every frozen event field, consumed the natural draw, and forced that one successful child gate to fail. Each branch was followed for 128 offsets after intervention or until first original-address death. Alternate child timing (same tick, delayed, or none) and address endpoint (handoff, lineage loss, or source-survival censoring) were recorded as independent axes.

3.4 Sequential retained-amplitude cut protocol

Finding 069 extended the committed Finding 068 cohort with a staged sequence of at most eight vetoes. At depth d, the complete selector prefix through $d-1$ was replayed from the common branch

state. For every branch that still handed off, the first next ancestry-preserving child was joined to its successful candidate-ledger row. Its tick and gate ordinal, source, target, candidate step and order, available amplitude, target state, probability, and random draw were written to a complete manifest and hashed and committed before the depth-d gate was vetoed. Every later replay had to reproduce all earlier selector identities and natural gate passes exactly. A veto consumed the frozen random draw but suppressed the transfer, thereby retaining at the source the amplitude that the natural realization would have spent.

Let $H_b^{(d)} = 1$ when branch b retains a collision-free descendant at first original-address death after the first d committed vetoes, and let $H_b^{(d)} = 0$ when that address dies without one. The measured depth is

$$D_b = \min\{d \in \{1, \ldots, 8\} : H_b^{(d)} = 0\}. \tag{8}$$

This is a prefix-minimal depth under the frozen first-next-child and retained-amplitude policy. It is not a minimum over arbitrary event subsets or state interventions. Every stage used the same endpoint window: 128 offsets after the depth-1 intervention. A depth-8 handoff was right-censored above the protocol maximum; an original address still active at the fixed horizon was separately source-survival censored. Collision, payload, or conservation failures were protocol failures, not post-treatment exclusions.

3.5 Independent ordered-cut replication

Finding 070 froze a 256-case screen from 32 previously unused initialization seeds before continuation outcomes. The lexicographically first strict eligible source per seed yielded four source geometries. Thirty-two factual branches per source were completed, and every collision-free natural continuation with exactly one inherited child was committed before any veto. The identical Finding 069 selector rule, retained-amplitude intervention, 128-offset endpoint, and depth-eight cap were then applied. Replication was preregistered as at least five eligible selectors per source and a depth-1 handoff in at least two newly selected sources; exact depth-1 matched shams and the full selector-hash chain remained required protocol checks.

4 Current Result: GC-001

In three controlled replay events across two seeds, payload-free Iota dynamics restored the same tracked connectivity relation through a causally necessary successor bridge having no cells in common with the predecessor bridge.

Of the 36 Class B events replayed, 12 had a role-bearing predecessor bridge and 3 passed the strict criterion. The positives occurred across 2 seeds. For all three positives, bridge-path Jaccard overlap was zero, reachability changed from one to zero to one, successor removal returned reachability to zero, and pre- and post-event shortest-path length was two. Payload was zero and the conservation ledger was exact.

Seed	Sample	Start	Rebridge	Predecessor	Successor	J
46	7	61	65	17:27:24	16:28:25	0.0
46	7	420	424	39:16:19	40:17:20	0.0
56	3	106	112	14:4:5	15:5:6	0.0

Table 1: Strict constraint-transport events in Finding 060. J is bridge-path Jaccard overlap.

4.1 Post-transport address persistence

Finding 061 followed the original predecessor addresses after rebridge. The three strict transports preserved their restored relation for 9, 3, 51 recorded offsets, whereas all 9 role-bearing failures had zero relation lifetime at nominal rebridge. Each strict-positive address lineage terminated when its singleton predecessor address became inactive. No second continuous strict transport was observed through offset 1024.

4.2 Prospective expression lineage

Finding 062 instrumented successful hard-transfer activations and verified that the observer left paired simulation and RNG states exactly unchanged across 94 observed transfers. Under the preregistered source-to-new-target ancestry rule, the three strict-event relation lifetimes were 170, 601, 51 offsets. 2 of the three singleton expressions continued after their original address became inactive; the longer remained alive at the offset-1024 censoring horizon. All nine role-bearing failures remained at zero lifetime, no lineage collisions occurred, and no second strict transport was observed.

4.3 State-conditioned handoff susceptibility

Finding 063 branched 32 continuation RNG streams from each of the three strict-rebridge states. Handoff rates were 50.0%, 90.6%, 87.5%, respectively. Two-sided Fisher exact comparisons gave $p = 0.000752$ for the first versus second seed-46 state, $p = 0.00251$ for the first seed-46 versus seed-56 state, and $p = 1.0$ for the two high-susceptibility states. All 96 original addresses deactivated within the declared horizon, no lineage collision occurred, payload remained zero, and the ledger remained exact.

4.4 Held-out compact-rule test

Finding 064 trained a preregistered decision stump on 48 terminal-aligned branches and evaluated it once on 48 held-out branches. The selected rule was singleton-lineage size greater than 1.5 immediately before original-address death. Held-out sensitivity was 0.361, specificity was 1.000, and balanced accuracy was 0.681 ($p = 0.0210$, two-sided Fisher exact). The marker had no false positives but missed 23 of 36 handoffs; it is therefore treated as a sufficient state marker, not a general continuation law.

4.5 Fresh confirmation of a static-geometry rule

Finding 065 excluded endpoints with preexisting descendants, selected a new stump using only the original discovery rows, and evaluated it on 96 fresh continuations. The frozen rule predicted child creation for at most one active cube-26 neighbor. Among 71 eligible single-address endpoints, held-out balanced accuracy was 0.582 and the association was not significant ($p = 0.240$, two-sided Fisher exact). The preregistered static terminal features therefore did not yield a confirmed general continuation rule.

4.6 Ordered terminal events and a causal continuation gate

Finding 066 instrumented every selected hard-transfer candidate on the terminal tick for 96 fresh branches. The observer passed exact state-and-RNG equivalence and recorded 20,748 terminal candidate events. The ordered ledger classified all 96 outcomes without collisions: 57 death-tick child

activations, 16 preexisting-descendant continuations, 19 source-exhaustion deaths, three transfer-gate deaths, and one target-preemption death. In a preregistered replay of one gate-failure branch, permitting exactly one failed successor gate while still consuming its original random draw changed the outcome from lineage death to one collision-free ancestry-preserving child. This supports a bounded causal claim: continuation can depend on realizing an eligible successor before the source expends its terminating amplitude.

4.7 Prospective causal replication and independent geometries

Finding 067 froze three one-decision policies before evaluating 192 fresh branches from the original source states. Permitting the first failed gate to an inactive successor produced an immediate inherited child in 6/6 cases. Vetoing the successful non-child transfer that would spend the final source amplitude reopened the route in 30/30 cases; three formed a child immediately and 28 eventually handed off. Exact matched shams, one-event matching, payload absence, conservation, and collision freedom held throughout.

A complete screen of 128 continuations from 16 unused initialization seeds found four singleton-eligible strict transports across two seeds and froze one primary source per seed before terminal outcomes. In 64 branches from these 2 independent geometries, seed 71 produced eight successor-gate deaths; all eight interventions produced immediate children and handoff. One seed-58 exhaustion intervention reopened the route but did not eventually hand off, while one seed-71 preemption intervention restored target availability and was followed by handoff four offsets later. These results separate two causal conditions: route availability and successor realization. Reopening opportunity is causal, but it is not by itself sufficient for continuation.

4.8 Necessity and temporal redundancy of realization

Finding 068 completed 256 fresh factual branches and committed 164 unique-child selectors before any veto outcome. Suppressing the selected child produced an alternate child on the same tick in 5 branches, after a delay in 117, and never in 42. All 122 branches with an alternate child handed off at first address death; all 42 branches without one lost the lineage. No source-survival endpoint was censored.

The selected target was the first alternate target in only 17 cases; 105 alternatives used a different address. Source capacity was retained immediately after veto in 161/164 branches. Both temporal redundancy and branch-level necessity occurred in every represented initialization seed. All selected events were suppressed, all shams and selector identities were exact through the original draw, and no collision, payload, or conservation failure occurred.

4.9 Sequential realization cut depth

Finding 069 began with the same 164 factual unique-child branches and froze each next-depth selector manifest before applying the newly selected veto. The observed cut counts at depths 1 through 8 were 42, 20, 22, 17, 11, 13, 13, and 5, respectively. Thus 143 branches reached first original-address death without a descendant under an observed policy prefix. After all eight vetoes, 19 branches still handed off and are right-censored above the protocol maximum; 2 additional branches retained the original source through the 128-offset window and are source-survival censored rather than assigned a cut depth.

Across the adaptive chains, 648 distinct branch-selector identities were committed. Replaying every complete prefix produced 2142 gate-override executions. 80 branches selected at least one target address that had appeared earlier in the same selector chain, showing that route replacement

Depth	Risk set	New cuts	Handoffs	Source-censored
1	164	42	122	0
2	122	20	102	0
3	102	22	80	0
4	80	17	63	0
5	63	11	52	0
6	52	13	39	0
7	39	13	26	0
8	26	5	19	2

Table 2: Finding 069 ordered realization cut outcomes. Each depth's risk set is the preceding depth's confirmed handoffs.

may revisit an address as well as move to a new one. All selector hashes, parent-chain hashes, branch nesting, inheritance joins, and event identities verified. There were 0 lineage collisions; payload remained zero and the conservation ledger remained exact in every stage. These counts estimate the ordered, retained-amplitude policy response, not a globally minimal successor cut.

4.10 Independent replication across initialization geometries

Finding 070 selected 4 new source geometries and froze 79 factual unique-child continuations. The observed depth-1 through depth-8 cut counts were 23, 15, 7, 9, 4, 3, 5, and 2, respectively. 68 branches had an observed cut, 10 still handed off after eight vetoes, and 1 was source-survival censored at depth seven.

Depth	Risk set	New cuts	Handoffs	Source-censored
1	79	23	56	0
2	56	15	41	0
3	41	7	34	0
4	34	9	25	0
5	25	4	21	0
6	21	3	18	0
7	18	5	12	1
8	12	2	10	0

Table 3: Finding 070 independent ordered-cut replication. Each depth's risk set is the preceding depth's confirmed handoffs.

Every source geometry produced at least one depth-1 handoff, exceeding the preregistered two-source criterion. The chains contained 286 committed selector events and 923 override executions; 38 branches reused a target. One observed cut contained a transient inherited child that disappeared before address death. All hash, join, sham, payload, conservation, and collision checks passed.

4.11 Causal pre-articulation transition geometry

Finding 074 screened 37 strict transports without counterfactual outcomes, found 35 eligible pre-articulation channels, and froze one channel in each of 25 independent initialization seeds. Suppressing the frozen precursor activation prevented the recorded later component split in all 25 channels. Moving the exact post-transfer state to a preregistered nearby depleted address also prevented the recorded split in all 25.

When activation and amplitude remained at the exact articulation address but inherited R, V, P, and angular memory were restored to their pre-transfer values, the split remained in 18 channels and was suppressed in 7. The complete preregistered pattern therefore occurred in 18 channels, exceeding the 17-channel threshold. Natural and sham trajectories, frozen candidate identities, intervention counts, payload absence, and conservation were exact in all 150 replays.

Vetoing the first natural successor-bridge activation extended split lifetime in 23 of 25 channels. However, 0 remained separated for 128 offsets. Thus the primary exact-placement gate passed while the separate durability extension failed.

4.12 Prospective test of a rooted graph-role representation

Finding 075 acquired 1,536 factual cases from 192 new initialization seeds, yielding 46 strict transports in 29 seeds. Outcome-sealed screening retained 45 eligible precursor channels and froze 25 independent common-donor topology channels. Sixteen common RNG branches per channel gave 400 outcomes for each of three variants and 1,200 branch outcomes overall.

Natural, graph-role-preserving, and graph-role-breaking placements produced 20, 24, and 14 articulations, respectively. Paired discordances favored preservation 21 to 11, but the preregistered one-sided exact McNemar test gave $p = 0.0551$ and failed the $p < 0.05$ gate. Preservation was higher in 8 of 11 informative channels, meeting that directional gate, but was strictly closer to the natural rate in only 5 channels versus 17 required. All factual controls, branch safety checks, state multisets, and swap invariants were exact. The composite graph-role hypothesis failed; Findings 076 and 077 were not opened.

4.13 Factorial successor-accessibility feasibility and prospective stop

Finding 078 separated structural placement from inherited state marks and enumerated a marked pre-gate successor graph. All 175 states in 25 development channels were conserved, payload-free, ledger-exact, RNG-neutral, and repeat-exact. All 25 channels were factorially identifiable, and 23 had a nonzero interaction in successor openness or repair affordance. The outcome-free selector chose 14 formerly preserving and 11 formerly breaking receivers. This was an instrument feasibility result, not an outcome confirmation.

Finding 079 then acquired 1536 factual continuations in a new seed range, obtaining 30 strict transports across 22 independent seeds. The precursor screen retained 25 eligible events in exactly 20 independent channels. The first common-donor failure reduced the maximum cohort below the preregistered minimum. The completed audit retained only 16 independent channels versus 20 required. The repair-interaction filter was not run, no branch manifest was created, and no outcome or directional effect was inspected.

5 Discussion

The strict positives show that a connectivity relation can, in a restricted event class, outlast complete replacement of the cells that instantiate its bridge. The particular bridge is therefore not sufficient to identify what persists in these events. The invariant candidate is the restored reachability relation between tracked region descendants.

This observation is consistent with the organizing hypothesis: interaction can alter the set of reachable collective states, and a constraint on that set can be re-instantiated in a different local realization. The result does not by itself validate the general reachable-set information measure in Equation 2.

Findings 066–067 add an event-resolved mechanism. A prospective expression can survive the death of its stationary address when an ancestry-preserving successor is realized while that route remains available. A lawful route may be lost because a gate fails, the source spends its final amplitude elsewhere, or another source activates the target first. The operational content of *before* is therefore causal rather than merely chronological: changing one identified decision changes whether the route remains available or whether a child is realized.

The intervention results also prevent a stronger but unsupported conclusion. Route availability and realization are distinct. Every exhaustion intervention in the prospective same-state cohort reopened the route, yet two did not later hand off; the independent seed-58 exhaustion case likewise reopened opportunity without rescue. Information, in the organizing language, concerns a structured set of still-possible futures whose members have lawful lifetimes. Persistence requires both an open route and timely realization. No payload transport is needed for the operational lineage used here.

Finding 068 shows that the naturally realized child is not generally unique in the reachable structure. Although every factual branch expressed exactly one child, veto exposed an alternate realization in 122 of 164 counterfactuals, usually at a different target and usually after a delay. The particular child was therefore temporally redundant in those branches. In the remaining 42, no alternate child appeared and the lineage died at address loss, establishing branch-level but-for necessity for the selected realization.

Finding 069 shows that this redundancy can recur through several ordered replacements. Observed policy cuts occurred at every tested depth from one through eight: 143 of 164 branches lost the lineage after some committed prefix, whereas 19 still handed off after all eight vetoes and 2 retained the original address at the fixed horizon. The depth is therefore a measure of robustness to this declared sequence of realization suppressions. It is not evidence that a static reserve contained that many interchangeable children. Each veto retained amplitude and changed the subsequent lawful path, on which a later successor could become available and be realized. Target reuse in 80 branches also reinforces the distinction between a stationary address and the lineage relation expressed through it.

Together these results support a sharper hierarchy. Information describes which futures remain possible; constraint determines which are lawful; ordered interaction gives those routes finite opportunity lifetimes; realization selects one route; and redundancy determines whether another realization can substitute when that selection is removed. The natural trajectory alone concealed most of this redundancy because it displayed one child even when a different future could become reachable along the retained-amplitude counterfactual path.

Finding 070 moves that conclusion beyond the original selected source states. Every one of four prospectively screened independent geometries admitted an alternate continuation after the first veto, and cuts again occurred at every tested depth. This supports geometric recurrence of ordered continuation redundancy under the frozen law and scheduler. It does not turn the adaptive chains into a static map: target reuse and retained amplitude mean that later routes can revisit addresses or arise only in the altered history.

Finding 074 directly changes a precursor event in the successor structure rather than fitting a scalar spatial representation. Across 25 independently initialized strict channels, both deleting the activation and relocating its post-transfer state prevented the same later articulation event. Exact address placement is therefore causally important for this frozen outcome. The seven memory-sensitive channels prevent a placement-only law: in those geometries, activation and amplitude at the address were insufficient without some member of the erased state bundle. The supported object is consequently a coherence-conditioned transition geometry with heterogeneous marked-state dependence, not a radial field or particle-like source.

The rebridge veto gives a complementary boundary. It usually prolonged the topological opening, but never produced the preregistered 128-offset durable separation. The result supports causal control over a successor-graph transition; it does not establish durable morphogenesis or a physical field-level correspondence.

Finding 075 then tested whether a frozen rooted local signature explained susceptibility to that transition. The paired outcomes contained a suggestive directional contrast, but the confirmatory probability gate narrowly failed and the independent natural-closeness gate failed strongly. The registered result is therefore negative. It rules out this bounded signature as a sufficient account under the frozen swap policy; it does not rule out relational topology. The next representation should preserve marked successor-event structure, larger relational context, or temporal order rather than treating the observed contrast as permission to tune the failed signature.

Finding 078 shows that structural placement and inherited marks can be separated operationally and can interact non-additively in a marked one-step successor representation. It does not show that this representation predicts later articulation. Finding 079 was designed to supply that independent test, but its support pipeline failed before outcome measurement. The correct status of state-conditioned repair affordance is therefore untested, not negative. This distinction preserves the positive causal placement result from Finding 074 while preventing developmental instrument behavior from becoming a confirmation claim.

5.1 Candidate bridge to Gravity from Entropy

The exact identity in Equation 4 supplies a nonverbal bridge from the current support ratio to relative entropy. It does not yet connect Iota dynamics to gravity. In GfE, the local Lagrangian is

$$\mathcal{L}_{\mathrm{GfE}}(x) = -\operatorname{Tr}_F \ln \mathbf{\Theta}_x, \qquad \mathbf{\Theta}_x = \widetilde{\mathbf{G}}_x \widetilde{g}_x^{-1}, \tag{9}$$

where $\widetilde{g}$ is the physical higher-order metric, $\widetilde{\mathbf{G}}$ is the metric induced by matter and curvature, and $\mathbf{\Theta}$ is required to be positive [4, 5]. These metrics are not normalized probability distributions. In the finite flattened representation, Equation 9 is a log-determinant contrast, $\mathcal{L}_{\mathrm{GfE}} = -\ln \det_F \mathbf{\Theta}$.

A concrete bridge theorem would require a positive future-law operator $\mathbf{Q}_x$ derived from the Iota transition structure such that

$$\begin{aligned} \mathbf{\Theta}_x &\overset{?}{=} (\mathbf{Q}_x^{\mathrm{prior}})^{-1/2} \mathbf{Q}_x^{\mathrm{post}} (\mathbf{Q}_x^{\mathrm{prior}})^{-1/2} \sim \mathbf{Q}_x^{\mathrm{post}} (\mathbf{Q}_x^{\mathrm{prior}})^{-1}, \\ \det_F \mathbf{Q}_x &= a\, \mu(\mathcal{R}_x), \qquad a > 0 \text{ common to prior and post.} \end{aligned} \tag{10}$$

Here $\sim$ denotes similarity, so the first representative is positive while retaining the determinant ratio of the original product. For the present finite Iota identity, μ is counting measure. A non-counting extension would define a new $\Delta I_x^{(\mu)}$ rather than silently replace the cardinality-based ΔI_x. Under these algebraic conditions,

$$\begin{aligned} \mathcal{L}_{\mathrm{GfE}}(x) &= \ln \frac{\mu(\mathcal{R}_x^{\mathrm{prior}})}{\mu(\mathcal{R}_x^{\mathrm{post}})} \\ &= \ln \frac{|\mathcal{R}_x^{\mathrm{prior}}|}{|\mathcal{R}_x^{\mathrm{post}}|} = \Delta I_x. \end{aligned} \tag{11}$$

The question mark is substantive: a physical bridge must still derive the operators, metric, covariance, variation, and continuum limit. Matching determinants after the fact would only reproduce a scalar.

GfE thermodynamics further writes a local mode contribution as $\delta s_k/\delta v = \ln \mathcal{G}_k$, so that $\mathcal{G}_k = \exp(\delta s_k/\delta v)$ and its intrinsic geometric temperature is $\theta_k = \mathcal{G}_k - 1$ [5]. If an Iota mode-resolved contribution $\chi_{x,k}$ and a physical mode dictionary were independently derived, the conditional sectorwise crosswalk would be

$$\begin{aligned} \chi_{x,k} &\widehat{=} c_k := \frac{\delta s_k}{\delta v}, \\ \mathcal{G}_k = e^{\chi_{x,k}}, \qquad \theta_k = e^{\chi_{x,k}} - 1, \qquad & \frac{\delta \epsilon_k}{\delta v} = e^{\chi_{x,k}} - 1 - \chi_{x,k}. \end{aligned} \tag{12}$$

Here $k \in \{0, \ldots, 4\}$ labels the five GfE sectors and z_k is the corresponding degeneracy, with $z_0 = z_1 = 1$ and $z_2 = z_3 = z_4 = 3$. In the reported weak-curvature Friedmann expansion, $u_k = \omega_k H^2$ with $\omega_k = \omega_k' \ell_{\mathrm{P}}^2$, and $\bar{\omega}_{[1]} := \sum_{k=0}^{4} z_k \omega_k$.

This cannot be imposed generally: individual GfE c_k can be signed, whereas Equation 3 is nonnegative. Nor does an aggregate $\chi_x \widehat{=} \sum_k z_k c_k$ determine a single temperature or energy, because $\delta\epsilon/\delta v = \sum_k z_k (e^{c_k} - 1 - c_k)$ depends on the full mode vector. The reference, normalization, mode dictionary, aggregation, and sign structure must be derived rather than selected for agreement.

For weak-curvature, late-time Friedmann backgrounds, modes with $\omega_k \neq 0$ obey $c_k \simeq \omega_k H^2$ and $\delta\epsilon_k/\delta v \simeq \frac{1}{2}\omega_k^2 H^4$. For non-de Sitter solutions with $H \sim t^{-1}$ these scale as t^{-2} and t^{-4}. After summation, the leading entropy coefficient $\bar{\omega}_{[1]}$ can vanish, including in the reported Milne and $w = 1$ cases, so an aggregate H^2 scaling requires $\bar{\omega}_{[1]} \neq 0$. Under the conditional mode map, an Iota quantity would inherit a scaling only in the corresponding nonvanishing sector. That is not yet a novel Iota prediction: it is inherited from the Friedmann approximation. Novel evidence would require an Iota-derived correction, transition, or invariant tied to measured succession variables such as opportunity lifetime or ordered cut depth, followed by a prediction that defeats ordinary branching, percolation, complete-state dynamics, and competing gravity models.

The bridge is therefore a precise research program. Construct $\mathbf{Q}_x$ from policy-indexed future laws; prove positivity, covariance, and robustness to scheduler, encoding, topology, and coarse-graining; recover the GfE metric operator and action at the functional-variation level; establish a controlled continuum limit; and only then derive an astrophysical or cosmological discriminator. The current simulator results supply a causal testbed for the first step, not evidence that the GfE action or gravity has emerged.

6 Limitations and Falsification Frontier

The structural foundation remains three strict Finding 060 events across two original seeds, all preserving a large-region/singleton relation. Finding 067's independent screen adds two primary initialization geometries, but only one contains a replicated gate-failure cohort of at least five cases. The independent exhaustion and preemption cohorts contain one case each and support only case-level mediator claims.

The ledger covers candidates selected by the declared hard-transfer routine; candidate selection and lexicographic source scheduling are not yet separately intervened upon and may be implementation-specific. Forcing a failed gate to pass is an efficacy test, not an estimate of natural probability. The original random draw is consumed and paired worlds are identical through intervention, but later RNG consumption can diverge after their states diverge.

Finding 068 conditions on factual natural continuation and therefore estimates its robustness, not the prevalence of necessity across all branches. Vetoing a successful child retains source amplitude and intentionally exposes alternate routes. Moreover, inherited realization is part of the

operational lineage definition, so the exact alternate-child/handoff correspondence is not an independent predictive association. Five source states represent four initialization seeds.

Finding 069 reports a prefix-minimal depth only for the frozen first-next-child policy. Because each suppressed transfer leaves amplitude at its source, successive counterfactual worlds do not expose a fixed list of routes present in the factual world. The result is therefore neither a graph-theoretic minimum cut nor evidence that every earlier member of a cutting prefix is individually necessary. The protocol also truncates two axes: 19 branches are right-censored above depth eight, and 2 are source-survival censored at the common 128-offset endpoint. Their cut depths are unknown.

Finding 070 independently replicates the policy response in four new source geometries, but its 79 branches remain conditional on factual unique-child continuation. Ten are depth-right-censored and one is source-survival censored; one cut includes a transient child that died before the endpoint. The replication therefore supports recurrence, not a scheduler-invariant topology, an entropy interpretation, or an exhaustive inventory of reachable futures.

Finding 074 is conditional on 25 strict singleton transports with an eligible depleted-target activation of the future articulation during the preceding 64 offsets. Its primary endpoint is the component split at the frozen natural offset, not prevention of every later separation. The memory intervention erases R, V, P, and angular memory as a bundle, so the necessary state coordinate in seven sensitive channels is unresolved. The transplant recipient was selected from its exact pre-step state, but ordinary update dynamics could change recipient memory on the intervention tick before the declared post-state move. Finally, rebridge veto never met the 128-offset durability gate.

Finding 075 is representation-specific. Its rooted signature preserves declared local counts, component counts, and degree sequence, but not graph isomorphism, exact external relational context, or event history. Ten of 25 channels produced no articulation in any variant, and branch RNG was reset after intervention to estimate conditional susceptibility rather than factual prevalence. The exact McNemar result was close to but above the frozen threshold and cannot be treated as positive; the separate natural-closeness requirement also failed decisively.

Finding 078 uses a completed development cohort and a radius-two, one-step candidate representation. Its nonzero factorial interactions establish neither an exhaustive reachable set nor downstream prediction. Finding 079 retained only 16 independent common-donor channels against a frozen minimum of 20. The nonzero repair-interaction filter and every factorial branch outcome were therefore left unopened; no effect estimate exists. The prospective hypothesis cannot be evaluated from this support failure.

Indefinite persistence, iterated strict transport, a necessary-and-sufficient continuation law, uniqueness to this update law, and physical correspondence remain unestablished. Further falsification should extend the frozen horizon and depth, compare alternative selector policies and source schedules, and enumerate small intervention subsets on held-out branches. Mass, energy, force, matter, and physical-field correspondence remain downstream hypotheses rather than demonstrated substrate primitives.

The relative-entropy result in Equation 4 is an exact mathematical identity under uniform nested supports, not an empirical validation of Equation 2. The reported experiments neither enumerate the complete multitime reachable set nor estimate the future laws in Equation 3. The proposed GfE operator map adds further untested requirements: a derived positive future-law operator, a metric and volume interpretation, sector and sign control, covariance, a continuum limit, and robustness to laws, schedulers, topologies, boundaries, and representations. No GfE scaling, gravitational transition, or observational signature has been derived or tested by the present finite simulator.

7 Reproducibility and Evidence

The paper uses generated rather than manually copied numbers. Canonical claims and definitions are in `evidence/claims.yaml` and `evidence/definitions.yaml`. Each `evidence/manifests/GC-NNN.json` records the exact path, byte count, and SHA-256 digest of supporting evidence and code; raw artifacts remain under `output/findings/`.

Finding 069's cases, prospective selector manifests, and aggregate tables are in `output/findings/finding069/`; its claim manifest identifies both implementations. Before reporting cut assignments, the synthesis verifies selector and parent-chain hashes, contiguous depths, nested risk sets, event identities, and safety invariants.

Finding 070's independent screen, factual freeze, eight staged risk sets, and replication synthesis are in `output/findings/finding070/`. Its synthesis revalidates the fixed source and selector hashes, nested risk sets, exact selector-to-case joins, shams, endpoints, and safety invariants.

`make verify` rebuilds both eight-stage syntheses, regenerates derived files, and validates expectations, provenance, paths, and freshness. `make paper` then compiles this PDF.

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